

## **Unnamed\_Unit**

### **Unit 3 - Creating Your Campfire**

#### **PDF Documents**

##### **PDF Document 1: 1738220607543-Chapter 5 - Unit 3 - Creating Your Campfire.pdf**

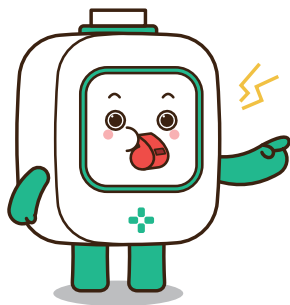
This PDF document is included in the unit materials.



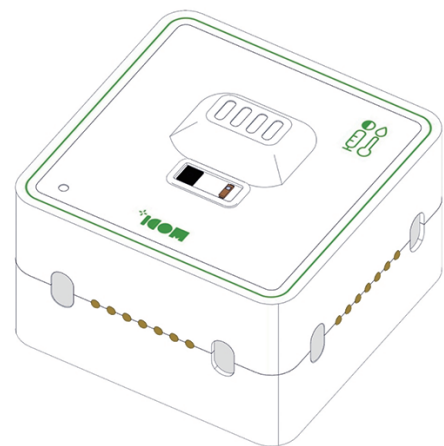
## UNIT 3 CREATING YOUR CAMPFIRE

Now that you know how to set up a campfire, it's time to get creative! In this activity, we'll make a pretend campfire that you can use indoors. This is a fun way to learn about campfires without needing any real fire.

**Do you remember what Environment do?**



It can sense four types of things!



**Environment**

### **TEMPERATURE**

Measures how hot or cold the surroundings

### **HUMIDITY**

Checks the amount of moisture in the air.

### **LIGHT**

Gauges the brightness of the light around it.

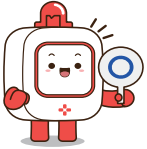
### **VOLUME**

Detects the level of sound in the environment.

## ● Introduction to Code Editor

### What is Code Editor?

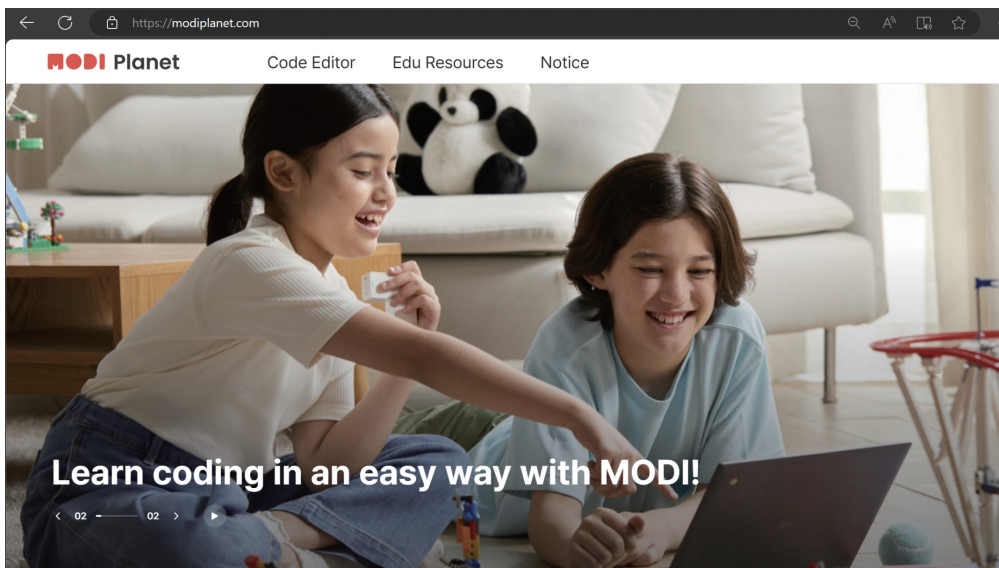
Code Editor is where we can programm MODI modules with block-based programming.



Let's get started by following the next steps:

#### Step 1

Go to 'MODI Planet' in your web browser. Here's the link: [www.modiplanet.com](https://modiplanet.com)



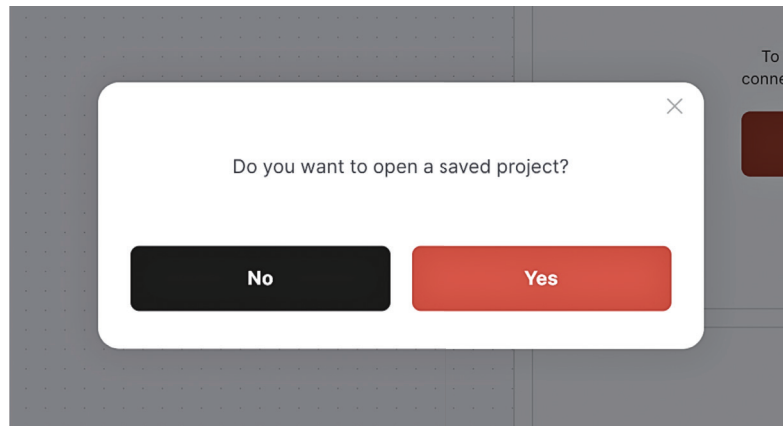
#### Step 2

Find 'Code Editor' in the top menu and click it.

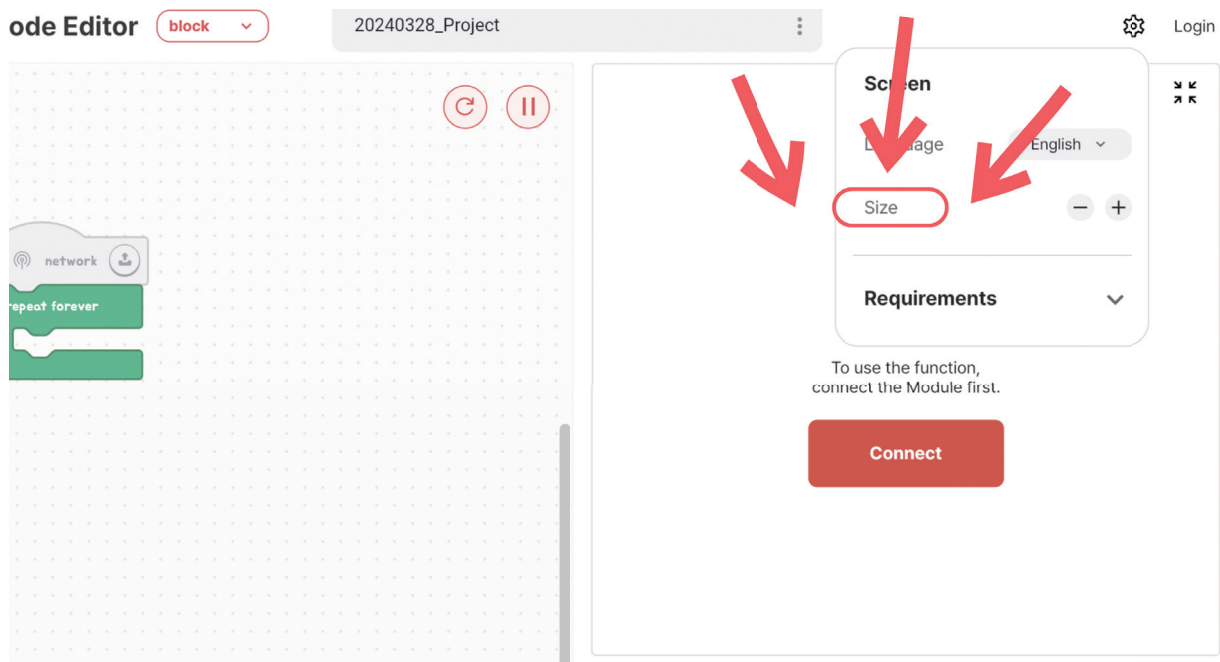




**Step 3** If it asks about opening a saved project, click 'No.'

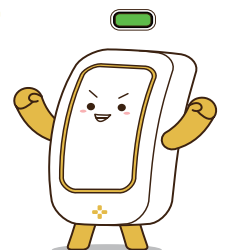


**Step 4** If the screen layout doesn't suit your preference, adjust the ratio by clicking the "Set" button.

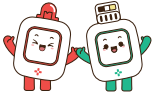


**TIPS**

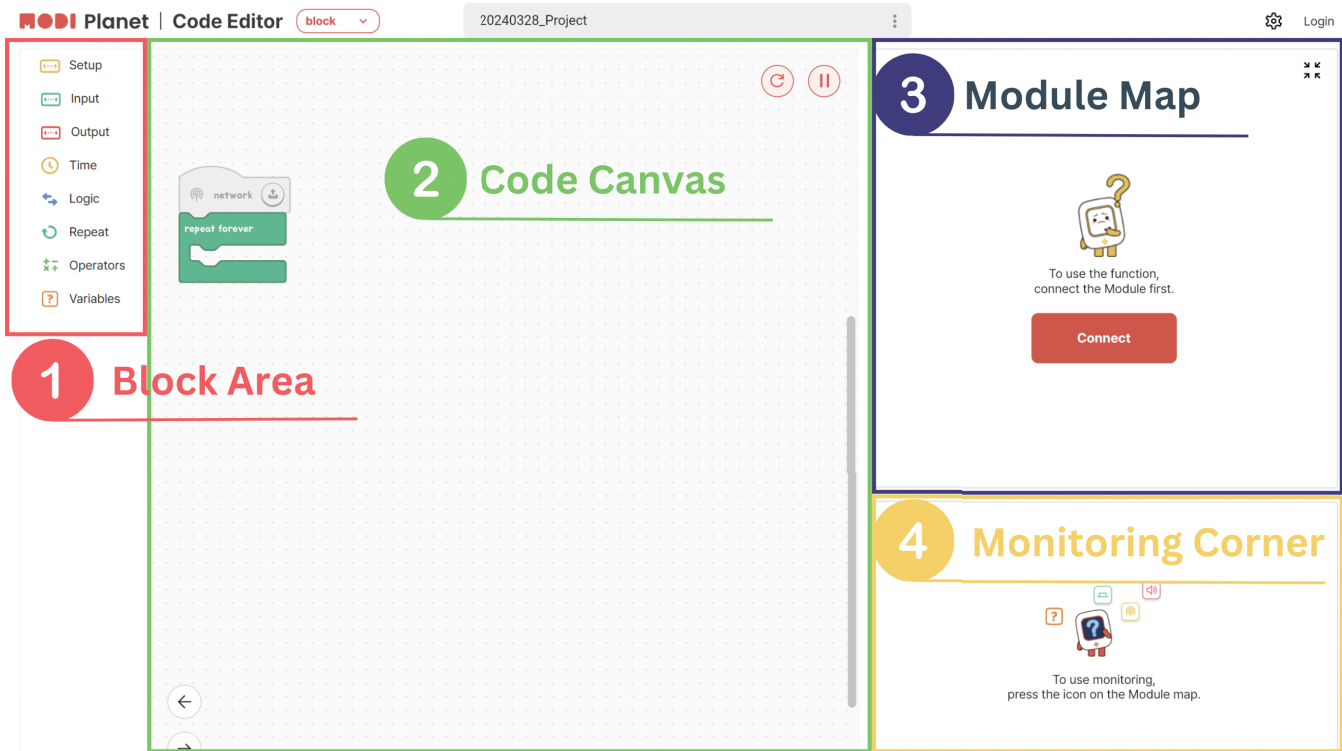
Depending on your screen size, some options might look different. If you see a button in the top right corner, use it to see more options.







## EXPLORING CODE EDITOR



### 1 Block Area :

This is where all the coding blocks live. They are sorted into eight groups based on what they do.

### 2 Code Canvas :

This is your space to build your code by putting blocks together from the Block Area.

### 3 Module Map :

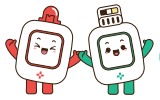
Here, you'll see icons for all the MODI modules connected to your PC. Clicking a module icon lets you check on that module.

### 4 Monitoring Corner :

This is where the monitoring screen appears. Depending on the size of your screen, it may appear as a pop-up window.

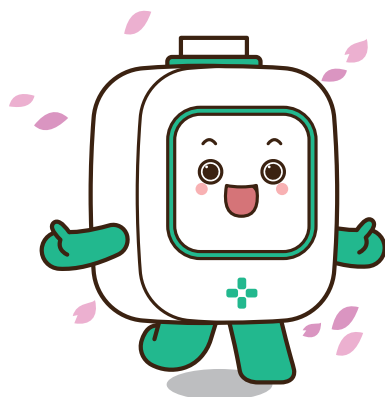
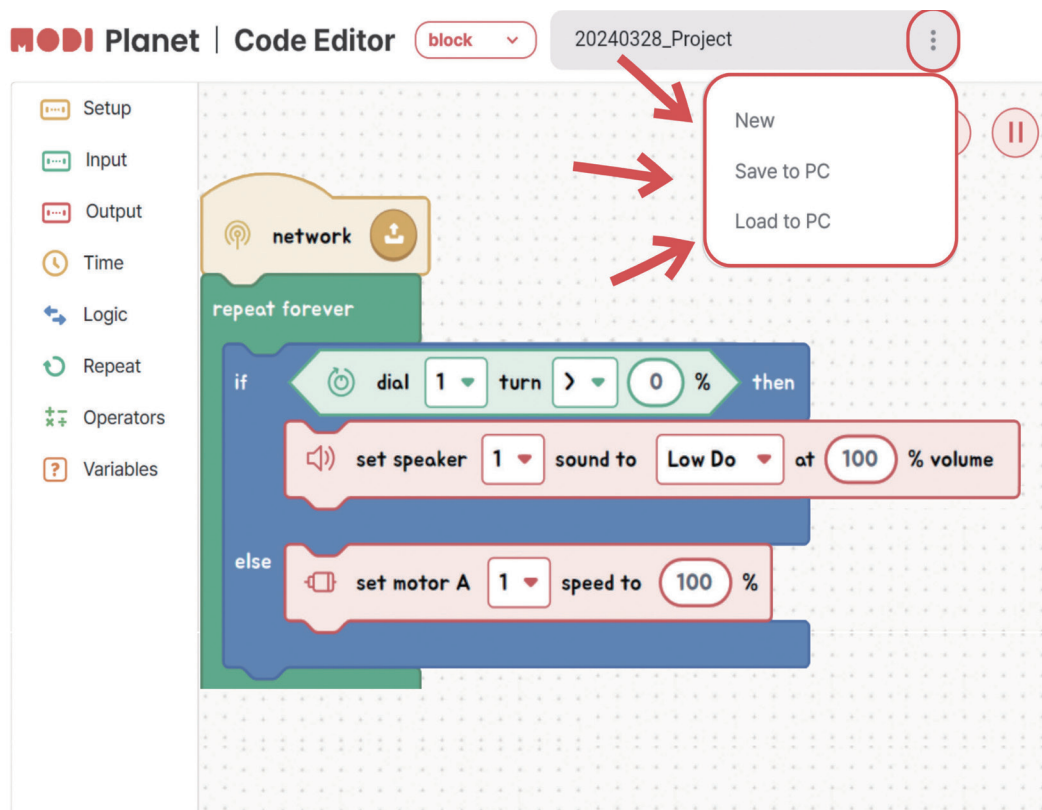


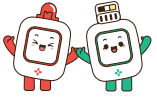
## INTRODUCTION TO CODE EDITOR



### SAVE & LOAD PROJECTS

Click the ellipsis in the picture below to open a popup with options to save and load your projects.

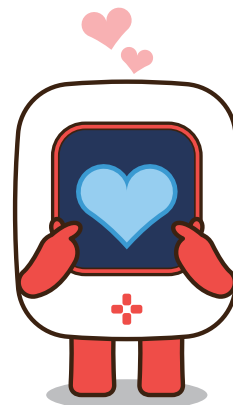




## INITIALIZING MODULES

- 1 Ensure the modules you wish to initialize are connected to your device.
- 2 Click the 'Initialization' button to reset them.

**MODI Planet | Code Editor** block 20240328\_Project



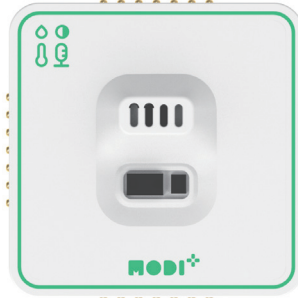


## Activity: Wishing Campfire

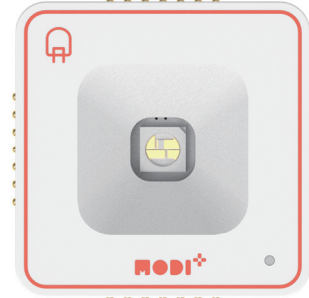
### 1. Materials Needed:



Network



Environment

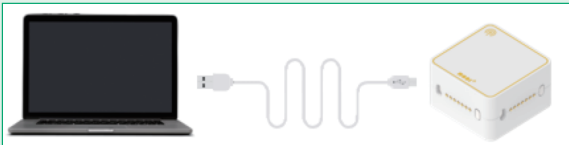


LEDs

### 2. Connect the Network with computer

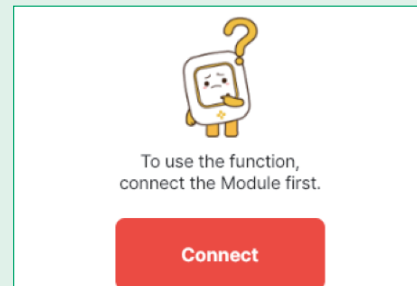
#### Step 1

Connect the Network Module to your device using a Type-C cable.



#### Step 2

Click the connect button in the module map.



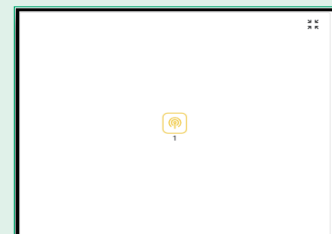
#### Step 3

Select 'MODI+ Network Module-Paired' and then click 'Connect' again.

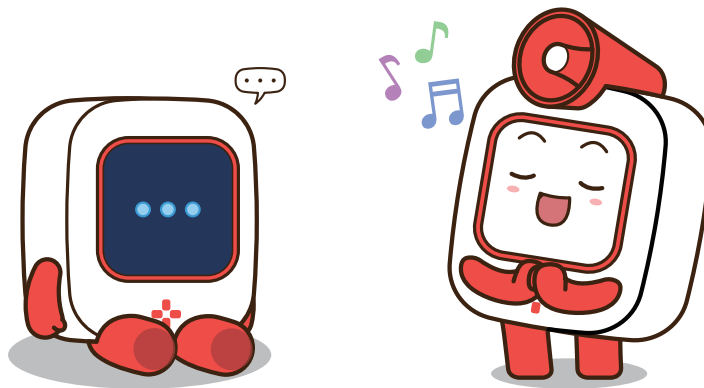


#### Step 4

A network icon will appear in the Module Map when the connection is successful.



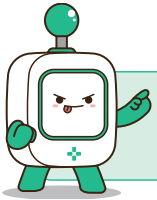
**3. Let's attach the Environment and LED modules next to the Network! Once everything is connected, you should see three modules displayed on the module map.**





## Coding

### Code Story



## Topic Camp Fire



### DEFINITION

### CODE DESIGN

We are on camp!  
Let's Light it up



### DECOMPOSITION

### STORY TO CODE

To make the campfire flames flicker or rise by blowing, what module would be needed?  
Please write down the names of the modules you can use.



### ABSTRACTION

### CODE DESIGN

How should it work?  
Shall we write down the functions in order?



Fire is off



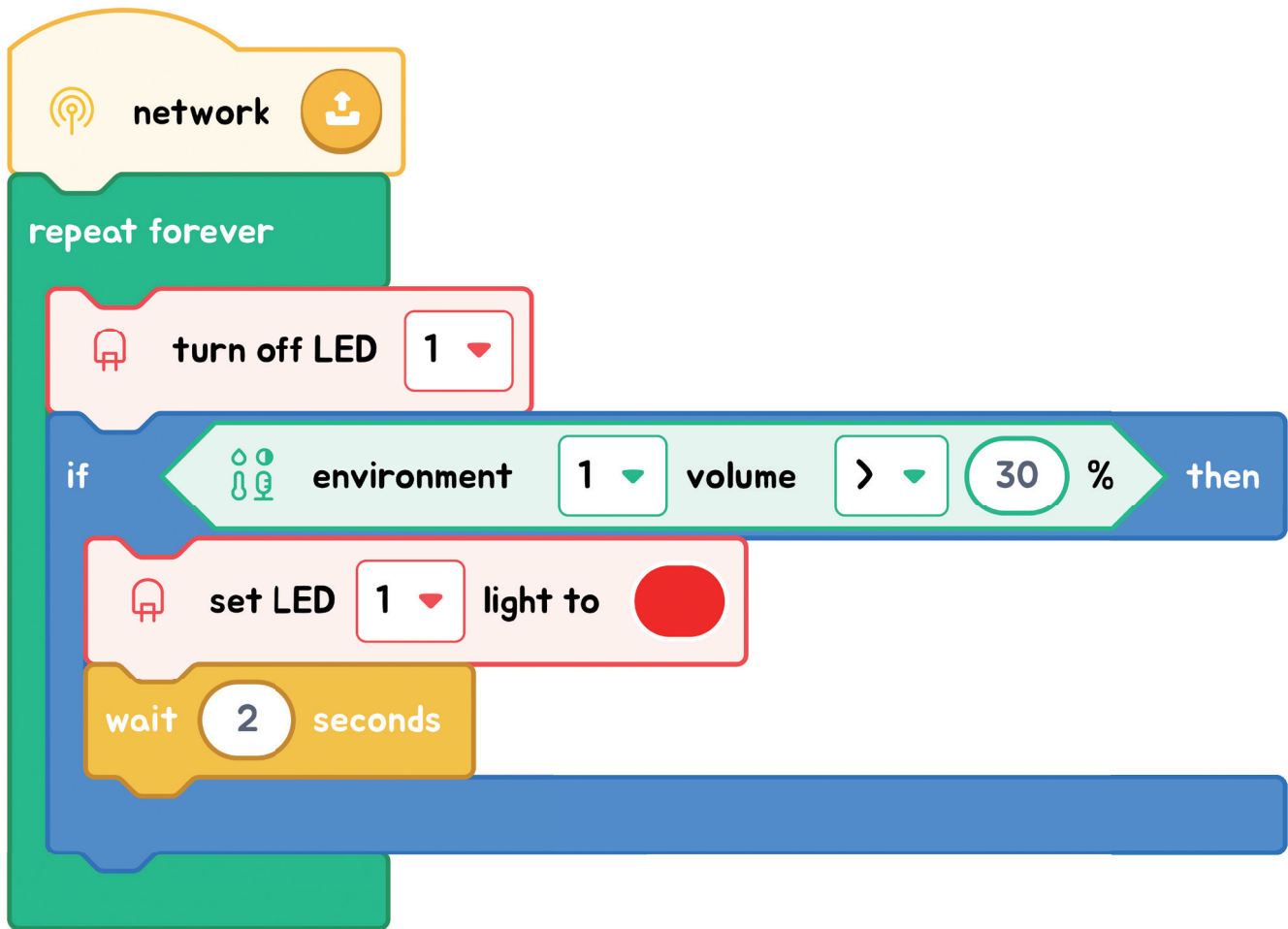
Blow the wood



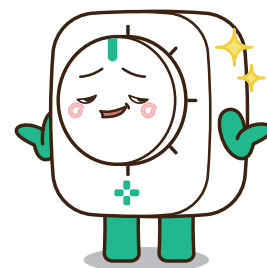
Fire is on for 2 seconds



Fire is off

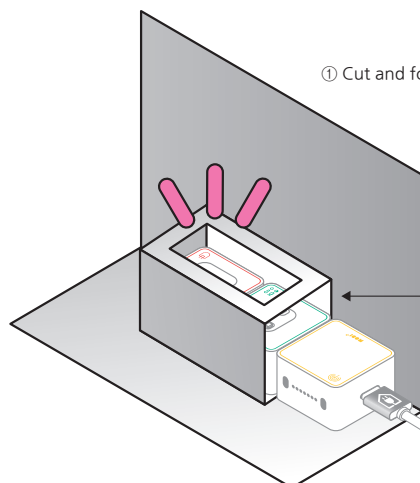
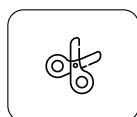


Now, let's use the template to create a camp fire!  
Combine the template and the modules well to make the fire.



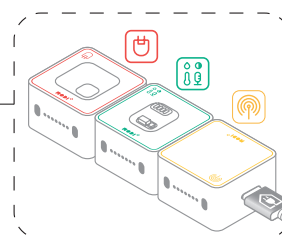


## Camp Frie



① Cut and fold like a pop-up card.

② Put the modules through it.





## [Guidelines]

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